SysML v2 Mermaid Viewer

Render the contents of a SysML v2 package as a Mermaid diagram (flowchart, class, state, sequence).

Version 0.0.1 · User Manual

Features

- Render the contents of a SysML v2 package as a Mermaid diagram from the editor (right-click → SysML Diagram).
- Five diagram types: flowchart, class, state, sequence and requirement.
- In-process parsing with the open-source SysML v2 parser (syside-languageserver); reusable profile libraries in sibling files are resolved automatically.
- Per-panel toolbar: ⟳ Refresh, ⤓ PNG export, and a syntax-error indicator.
- Auto-refresh open diagrams when the .sysml file is saved (configurable).
- Type-based styling from a JSON style file: colour elements by native SysML type and/or by the SysML type they conform to (hierarchy-followed).
- Ready-made Arcadia and NAF v4 styling profiles.
- Open several diagrams at once, one per panel.

Highlights

- **Flowchart.** Action flows with decisions, fork/join/merge and guarded transitions.
- **Class / State / Sequence.** Three further diagram types sharing the parser and rendering infrastructure.
- **Auto-refresh & PNG export.** Live refresh on save and one-click PNG export of any diagram.
- **Type-based styling.** JSON style files, with reusable Arcadia and NAF v4 profiles.

Examples

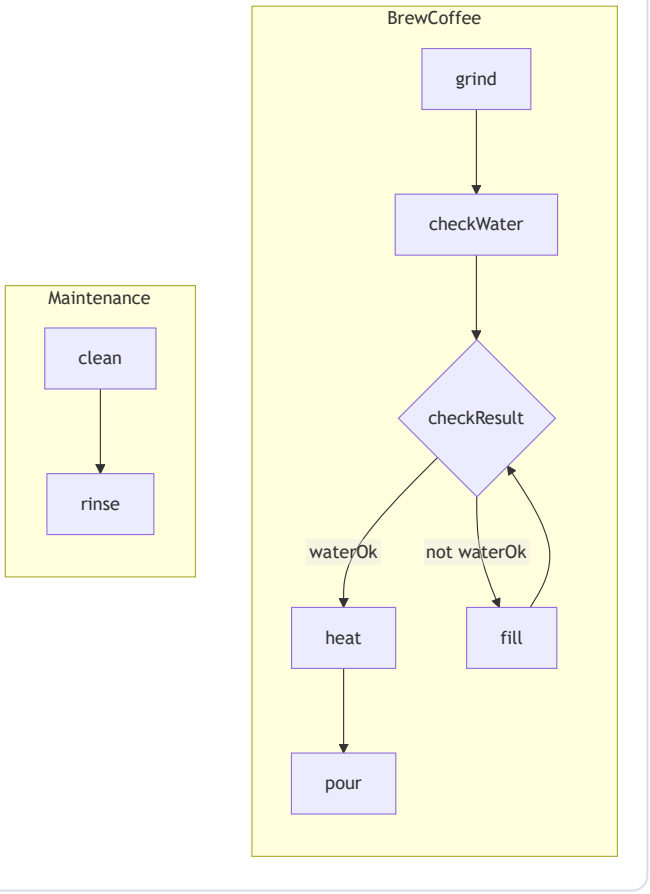
Flowchart — action flow with a decision

An `action def` with a decision node, guarded branches and a loop. Action usages become nodes, successions become edges, and transition guards label the edges. Non-flow elements (attributes, parts) are filtered out.

EXAMPLES/COFFEE.SYSML

```
package CoffeeMachine {

    private import ScalarValues::*;

    action def BrewCoffee {
        attribute waterOk : Boolean;

        action grind;
        action checkWater;
        decide checkResult;
        action heat;
        action fill;
        action pour;

        succession first grind then checkWater;
        succession first checkWater then
checkResult;
        first checkResult if waterOk then heat;
        first checkResult if not waterOk then fill;
        succession first fill then checkResult;
        succession first heat then pour;
    }

    action def Maintenance {
        action clean;
        action rinse;
        succession first clean then rinse;
    }

    part def Tank;
    attribute level : ScalarValues::Integer;
}
```

FLOWCHART DIAGRAM



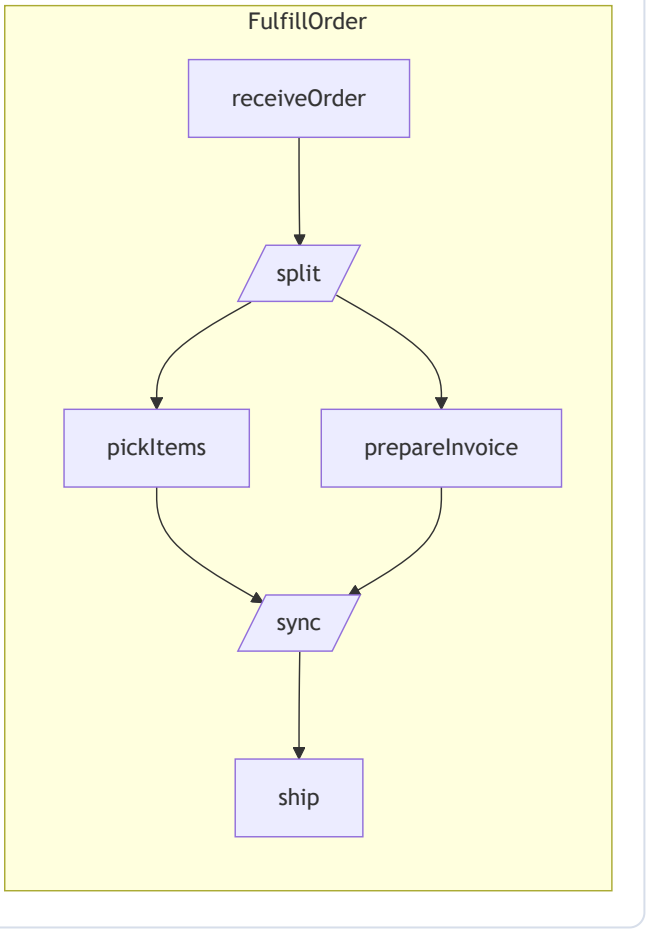
Flowchart — parallel actions (fork / join)

`fork` and `join` control nodes split work into parallel branches and re-synchronize them.

EXAMPLES/ORDER_FULFILLMENT.SYSML

```
// Flowchart example: parallel actions with fork /
join.
package OrderFulfillment {

    action def FulfillOrder {
        action receiveOrder;
        fork split;
        action pickItems;
        action prepareInvoice;
        join sync;
        action ship;

        succession first receiveOrder then split;
        succession first split then pickItems;
        succession first split then prepareInvoice;
        succession first pickItems then sync;
        succession first prepareInvoice then sync;
        succession first sync then ship;
    }
}
```

FLOWCHART DIAGRAM



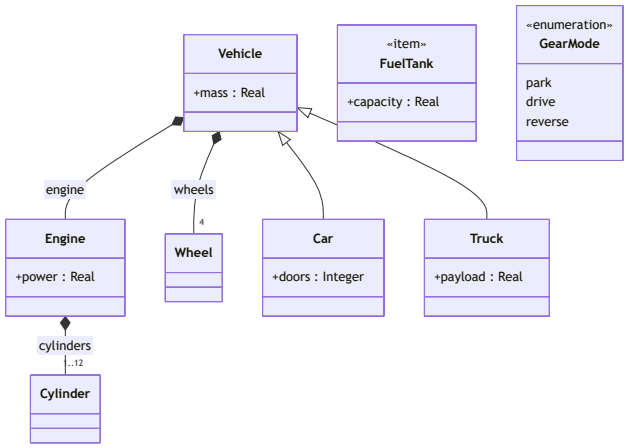
Class — structure, inheritance & composition

Definitions become classes; specializations (`:>`) are inheritance, typed part usages are compositions with multiplicities, and enumerations list their literals.

EXAMPLES/VEHICLE.SYSML

```
package VehicleModel {

    part def Vehicle {
        attribute mass : ScalarValues::Real;
        part engine : Engine;
        part wheels : Wheel[4];
    }

    part def Engine {
        attribute power : ScalarValues::Real;
        part cylinders : Cylinder[1..12];
    }

    part def Cylinder;
    part def Wheel;

    part def Car :> Vehicle {
        attribute doors : ScalarValues::Integer;
    }

    part def Truck specializes Vehicle {
        attribute payload : ScalarValues::Real;
    }

    item def FuelTank {
        attribute capacity : ScalarValues::Real;
    }

    enum def GearMode {
        park;
        drive;
        reverse;
    }
}
```

CLASS DIAGRAM



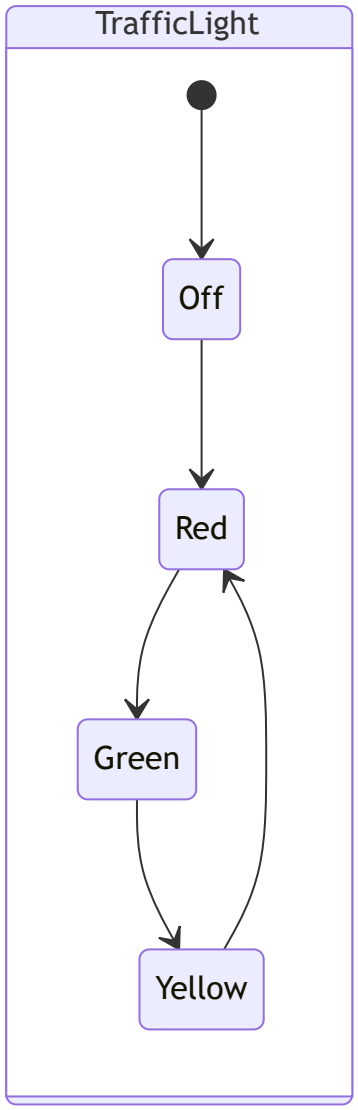
State — a traffic-light machine

A `state def` with its states and transitions. The `entry` pseudostate is rendered as the initial transition `[*]`.

EXAMPLES/TRAFFIC_LIGHT.SYSML

```
package TrafficLightSM {

    state def TrafficLight {
        entry; then Off;

        state Off;
        state Red;
        state Green;
        state Yellow;

        transition first Off then Red;
        transition first Red then Green;
        transition first Green then Yellow;
        transition first Yellow then Red;
    }
}
```

STATE DIAGRAM



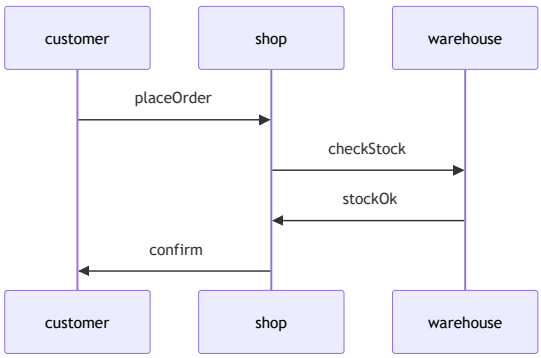
Sequence — message flow

Part usages become participants and directed `flow`s become ordered messages between them.

EXAMPLES/ORDER_PROTOCOL.SYSML

```
package OrderProtocol {

    part def Customer;
    part def Shop;
    part def Warehouse;

    part customer : Customer;
    part shop : Shop;
    part warehouse : Warehouse;

    flow placeOrder from customer to shop;
    flow checkStock from shop to warehouse;
    flow stockOk from warehouse to shop;
    flow confirm from shop to customer;
}
```

SEQUENCE DIAGRAM



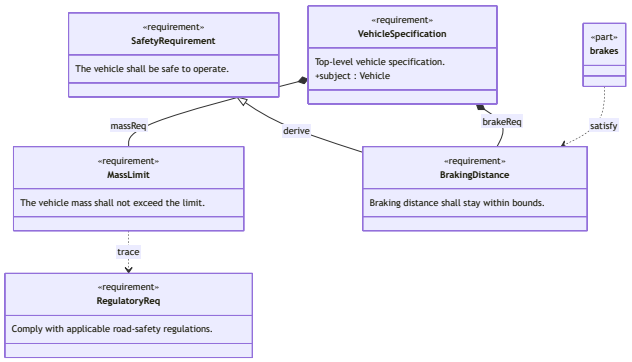
Requirement — derive, decompose & satisfy

Requirement definitions become «requirement» boxes. Specialization is derivation, nested requirement usages are decomposition, and a part that `satisfy`s a requirement is linked with a satisfy dependency. The subject is shown inside the box.

EXAMPLES/VEHICLE_REQUIREMENTS.SYSML

```
package VehicleRequirements {
    private import ScalarValues::*;

    part def Vehicle;
    part def Brakes;

    requirement def RegulatoryReq {
        doc /* Comply with applicable road-safety
regulations. */
    }

    requirement def SafetyRequirement {
        doc /* The vehicle shall be safe to
operate. */
    }

    requirement def MassLimit {
        doc /* The vehicle mass shall not exceed
the limit. */
        attribute maxMass : Real;
    }

    requirement def BrakingDistance :>
SafetyRequirement {
        doc /* Braking distance shall stay within
bounds. */
    }

    requirement def VehicleSpecification {
        doc /* Top-level vehicle specification. */
        subject vehicle : Vehicle;
        requirement massReq : MassLimit;
        requirement brakeReq : BrakingDistance;
    }

    // trace: the mass limit traces to a regulatory
requirement
    dependency from MassLimit to RegulatoryReq;

    part brakes : Brakes {
        satisfy BrakingDistance;
    }
}
```

REQUIREMENT DIAGRAM



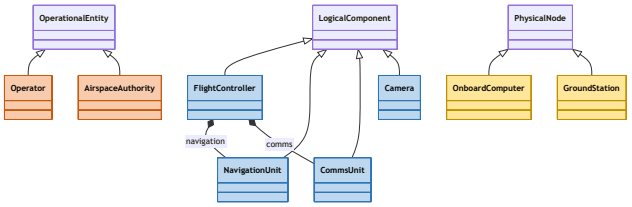
Styling — Arcadia colour code

With the Arcadia style file selected, components are coloured by their Arcadia type, following the specialization hierarchy: operational entities orange, logical components blue, physical nodes yellow.

EXAMPLES/ARCADIA/DRONE_SYSTEM.SYSML

```
// Arcadia-styled example: a small drone
surveillance system.
// View as a Class diagram with
sysmlMermaid.styleFile pointing at
// examples/arcadia/arcadia.style.json to get the
Arcadia colour code:
//   LogicalComponent → blue, PhysicalNode →
yellow, OperationalEntity → orange,
//   functions (actions) → green.
package DroneSystem {
    private import ArcadiaProfile::*;

    // --- Operational entities (orange) ---
    part def Operator :> OperationalEntity;
    part def AirspaceAuthority :>
OperationalEntity;

    // --- Logical components (blue) ---
    part def FlightController :> LogicalComponent {
        part navigation : NavigationUnit;
        part comms : CommsUnit;
    }
    part def NavigationUnit :> LogicalComponent;
    part def CommsUnit :> LogicalComponent;
    part def Camera :> LogicalComponent;

    // --- Physical nodes (yellow) ---
    part def OnboardComputer :> PhysicalNode;
    part def GroundStation :> PhysicalNode;

    // --- Logical functions (green) ---
    action def Stabilize :> LogicalFunction;
    action def StreamVideo :> LogicalFunction;

    // --- System assembly ---
    part drone : FlightController {
        part eye : Camera;
    }
    part computer : OnboardComputer;
    part ground : GroundStation;
    part pilot : Operator;
}
```

CLASS DIAGRAM



Styling — NAF v4 by grid layer

With the NAF style file selected, elements are coloured by NAF grid layer: Concepts purple, Service blue, Logical green, Physical/Resource orange.

EXAMPLES/NAF/C2_SYSTEM.SYSML

```
// NAF v4 styled example: a small Command & Control
architecture spanning the
// NAF grid layers. View as a Class diagram with
the NAF style file selected
// (SysML Mermaid: Select Style File… →
examples/naf/naf.style.json):
//   Capability → purple, Service → blue, Logical
→ green, Physical → orange.
package C2System {
    private import NafProfile::*;

    // --- Concepts layer: capabilities (purple) --
-
    part def SituationalAwareness :> Capability;
    part def CommandAndControl :> Capability;

    // --- Service layer (blue) ---
    part def TrackingService :> Service;
    part def MessagingService :> Service;

    // --- Logical layer (green) ---
    part def SensorNode :> LogicalNode;
    part def CommandPost :> LogicalNode;

    // --- Physical / Resource layer (orange) ---
    part def Radar :> PhysicalResource;
    part def Server :> PhysicalResource;

    // --- Architecture assembly ---
    part c2 : CommandAndControl {
        part awareness : SituationalAwareness;
    }
    part tracking : TrackingService;
    part messaging : MessagingService;
    part post : CommandPost {
        part sensors : SensorNode[*];
    }
    part radar : Radar;
    part server : Server;
}
```

CLASS DIAGRAM